Groundbreaker Solutions LLC Response to Sources Sought N61331-26-SN-Q18 Evaluation-Driven AI Capability Statement

1. Intent

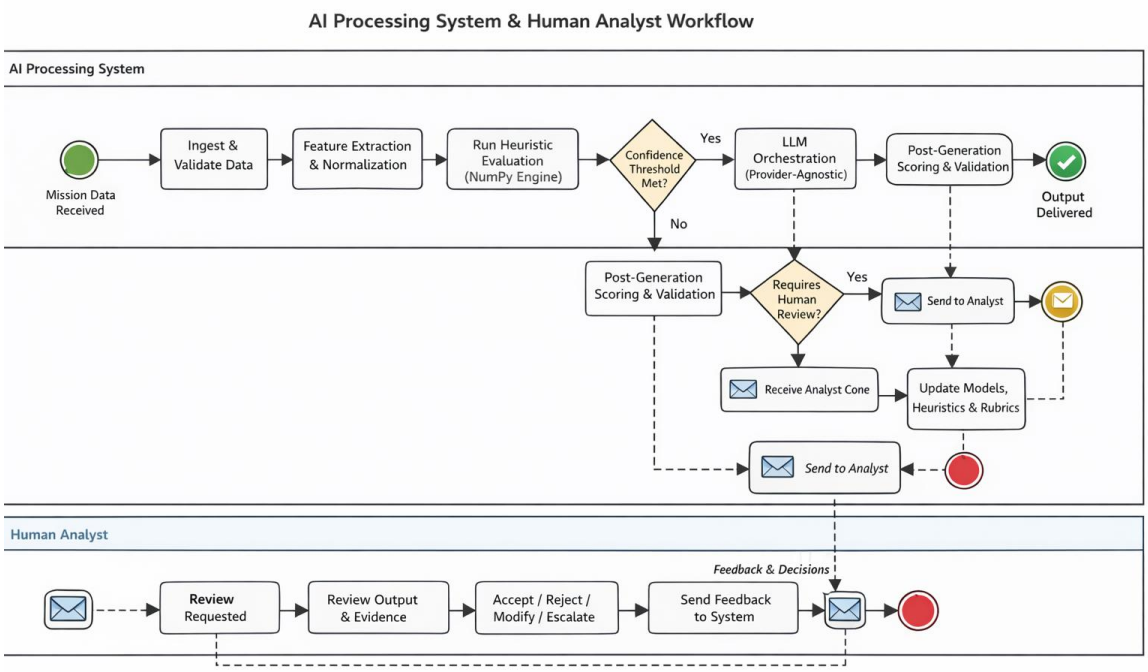
Groundbreaker Solutions LLC submits this capability statement in response to N61331-26-SN-Q18. Groundbreaker specializes in evaluation-driven, mission-aligned artificial intelligence systems designed to accelerate analyst workflows while maintaining strict governance, traceability, and operational reliability. Our approach integrates commercial, open-weight, and government-hosted models with selectively developed mission-specific models, governed by a human-in-the-loop architecture that enforces benchmark-driven performance and controlled deployment. We focus on delivering production-ready capabilities that can be sustained in operational environments rather than prototype-only systems.

2. Technical Positioning

Groundbreaker Solutions integrates commercial, open-weight, and government-hosted AI models with selectively developed mission-specific models to support advanced analytic domains, including cognitive influence and information environment assessment. By combining adaptable model orchestration with targeted model development, we enable rapid capability deployment while addressing specialized mission requirements. All capabilities are governed by an evaluation-first, human-in-the-loop architecture that enforces benchmark-driven performance, transparent scoring, and controlled operational integration.

This approach ensures that AI systems are not treated as standalone reasoning engines, but as components within a governed analytic workflow where outputs are measurable, explainable, and aligned to mission objectives.

3. Architecture Overview



The proposed architecture is an evaluation-driven AI platform that separates orchestration, evaluation, and model execution into modular layers. This separation enables provider-agnostic model integration, controlled evaluation, and scalable deployment across mission environments.

The architecture is designed around three principles: measurable performance, controlled integration, and extensibility across mission threads. Every system output is evaluated against defined benchmarks and confidence thresholds prior to operational use. Human analysts are incorporated into the workflow to ensure that edge cases and high-impact outputs are reviewed and adjudicated.

This modular design enables rapid adaptation to new data sources, analytic tasks, and model capabilities without requiring fundamental changes to the underlying system.

4. System Architecture Components

4.1 Orchestration Layer

The orchestration layer manages the end-to-end workflow, including data ingestion, evaluation coordination, model invocation, and output routing. A provider-agnostic model gateway enables integration with commercial platforms such as OpenAI and Grok, as well as government-hosted and locally deployed models.

Policy-based routing determines model selection based on mission constraints, including classification level, latency requirements, and task-specific considerations. The orchestration layer enforces structured input and output formats to ensure consistency and reliability across model interactions.

Failure handling and fallback strategies are built into the orchestration layer, enabling the system to recover from model errors or degraded performance by rerouting requests or invoking alternate models. All interactions are logged to support auditability and traceability.

4.2 Evaluation Layer

The evaluation layer is implemented using a custom Python/NumPy-based heuristic engine. This component performs deterministic scoring, benchmark alignment, anomaly detection, and confidence estimation across mission-relevant features.

Evaluation occurs both before and after model invocation. Pre-generation evaluation assesses data quality, completeness, and potential inconsistencies. Post-generation evaluation validates outputs against established rubrics, detects contradictions, and assigns confidence scores that inform routing decisions.

4.2.1 Evaluation Methodology

The evaluation methodology combines rule-based heuristics, statistical scoring, and LLM-as-judge techniques to assess model performance. Benchmark-driven scoring aligns outputs with mission-defined success criteria, while confidence scoring provides a quantitative measure of output reliability.

The use of transparent, vectorized computations enables repeatable and auditable evaluation processes that can be inspected and refined over time.

4.2.2 Cold-Start Evaluation

Cold-start evaluation capabilities enable the system to assess model performance in the absence of large labeled datasets. Heuristic rules and synthetic evaluation techniques provide initial performance baselines, allowing rapid iteration and refinement of models and workflows.

This approach supports early-stage deployment and reduces the time required to operationalize new capabilities.

4.2.3 Continuous Evaluation

Continuous evaluation mechanisms track performance metrics over time, including benchmark scores, confidence distributions, and analyst agreement rates. Drift detection identifies changes in model behavior, enabling timely updates to models and evaluation logic.

This ensures that the system remains aligned with mission requirements as conditions evolve.

4.3 Human-in-the-Loop Layer

Human analysts are embedded directly into the workflow as active participants in the evaluation process. Analyst review workflows are triggered based on confidence thresholds and mission impact considerations.

Review queues are prioritized to optimize analyst workload, ensuring that high-impact and low-confidence outputs receive attention. Analysts provide adjudication through structured actions, including acceptance, rejection, modification, and escalation.

Feedback is captured and used to refine heuristic models, prompt templates, and evaluation criteria, enabling continuous improvement and alignment with operational needs.

5. Operational Workflow

The system follows a structured workflow that ensures controlled processing and evaluation of all data and model outputs. The workflow includes data ingestion, feature extraction, heuristic evaluation, model invocation, post-generation validation, and confidence-based routing.

Decision points are embedded throughout the workflow to enforce governance and ensure that outputs meet defined performance criteria before being delivered to analysts or downstream systems. Human review is incorporated where necessary to validate outputs and provide feedback for system refinement.

This workflow supports auditability, traceability, and consistent operational performance.

6. MCP-Enabled Integration Strategy

The architecture leverages a standards-based integration layer, such as the Model Context Protocol (MCP), to expose mission services including heuristic evaluation, data access, and workflow actions. MCP standardizes access to these services, enabling modular integration and portability across environments.

Model execution is managed through a separate provider-agnostic gateway, ensuring that the system can adapt to evolving model capabilities without requiring changes to the underlying workflow or evaluation framework.

7. Data Ingestion and Processing

The system supports mission-aligned data ingestion across structured and unstructured sources. Data pipelines include source vetting, validation, normalization, and provenance tracking to ensure data integrity and traceability.

Feature extraction processes transform raw data into structured representations suitable for evaluation and model processing. These features support both deterministic scoring and contextual reasoning, enabling robust analysis across diverse data types.

8. Model Strategy

Groundbreaker integrates commercial and open-weight models, including OpenAI and Grok, alongside government-hosted solutions. Custom models are selectively developed using lightweight NumPy-based approaches to support specialized tasks such as classification, ranking, and domain-specific analysis.

This hybrid strategy enables rapid capability deployment while maintaining flexibility and control over mission-critical functions.

9. Deployment and Productionization

The architecture supports deployment across cloud, on-premises, and air-gapped environments. Components are containerized and integrated with CI/CD pipelines to ensure consistent deployment and rapid iteration.

Security considerations include data isolation, access control, and audit logging to support deployment in sensitive environments. Monitoring and observability capabilities enable performance tracking, fault detection, and system optimization.

10. Sustainment and Transition

Groundbreaker provides a structured sustainment approach that includes continuous evaluation, model updates, heuristic refinement, and analyst training. Performance metrics are tracked over time to identify areas for improvement and ensure alignment with mission objectives.

Transition plans include comprehensive documentation, onboarding guidance, and training support to ensure seamless integration into operational environments. Long-term support includes maintenance, updates, and ongoing optimization of system components.

11. Alignment to ONI Requirements

Groundbreaker's architecture aligns with ONI requirements by supporting iterative model development, benchmark-driven evaluation scoring, cold-start evaluators, mission thread extensibility, production-ready deployment packages, and mission-aligned data ingestion.

The system provides integration support, training, technical documentation, and maintenance capabilities, ensuring full lifecycle delivery of AI systems in operational environments.

12. Relevant Experience

Groundbreaker Solutions has delivered production-grade systems supporting real-time data processing and AI-driven workflows. The AMPS USSF Evolution Project supported mission-critical weather balloon instrumentation using ASP.NET MVC, SignalR, and PostgreSQL, addressing operational requirements in real time.

Programming.Team developed an LLM-agnostic resume platform that scaled to thousands of users, demonstrating the ability to orchestrate AI models across diverse use cases.

Courseware.Coach implemented AI-driven instructional systems using Azure Bot Framework and OpenAI technologies, delivering interactive learning environments that demonstrate advanced human-AI interaction capabilities.

13. Key Differentiators

Groundbreaker's approach is distinguished by its evaluation-first architecture, human-in-the-loop integration, provider-agnostic model orchestration, and production-ready design. The separation of evaluation, orchestration, and model execution enables flexibility, transparency, and operational reliability.

The use of MCP for service integration and a separate model gateway for inference ensures modularity and reduces vendor lock-in, while custom heuristic evaluation provides measurable performance and auditability.

14. Administrative Information

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Subcontracting: Groundbreaker does not intend to subcontract and will comply with FAR 52.219-14 requirements.